

Fuzzy Inference Systems

Fuzzy Inference Systems

Fuzzy inference (reasoning) is the actual process of ***mapping from a given input to an output using fuzzy logic.***

The process involves all the pieces that we have discussed in the previous sections:
membership functions, fuzzy logic operators, and if-then rules

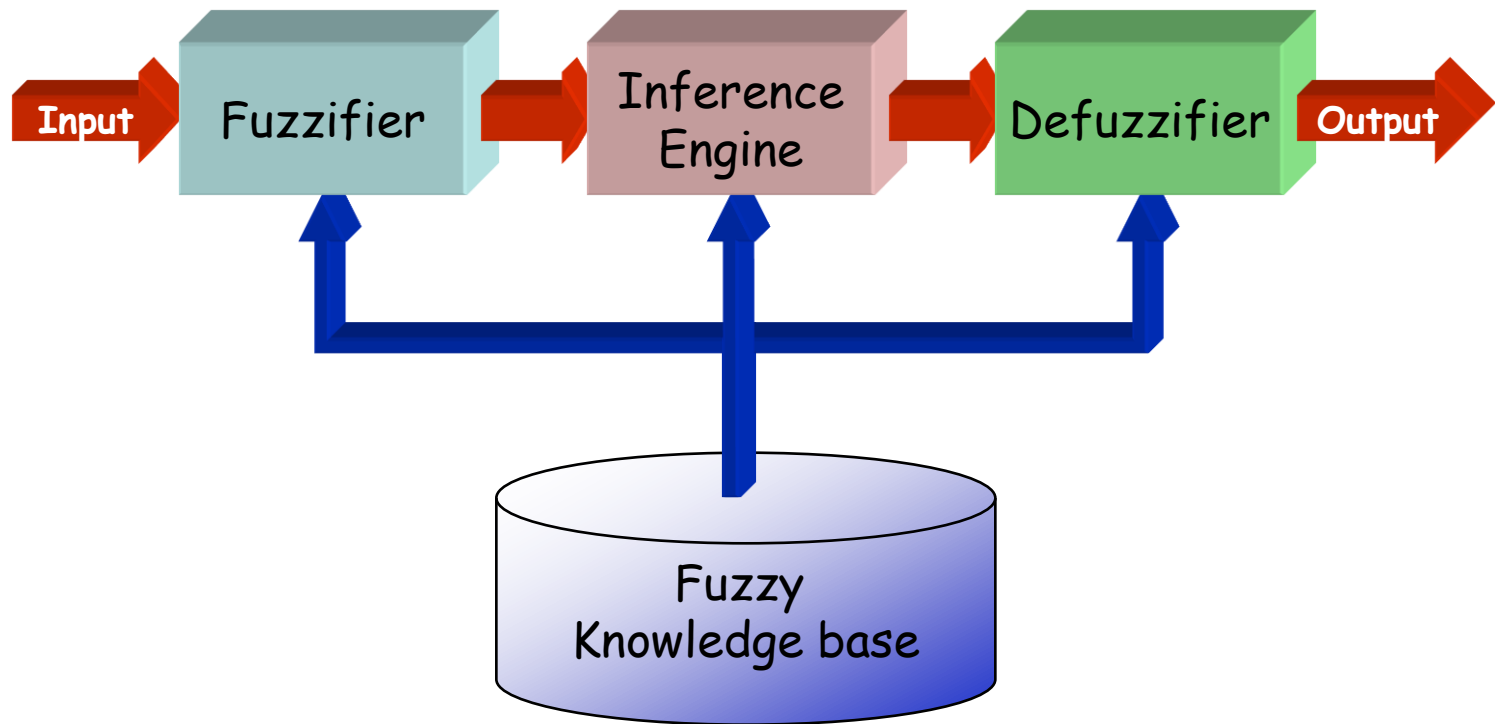
Fuzzy Inference Systems

Fuzzy inference systems have been successfully applied in fields such as automatic control, data classification, decision analysis, expert systems, and computer vision.

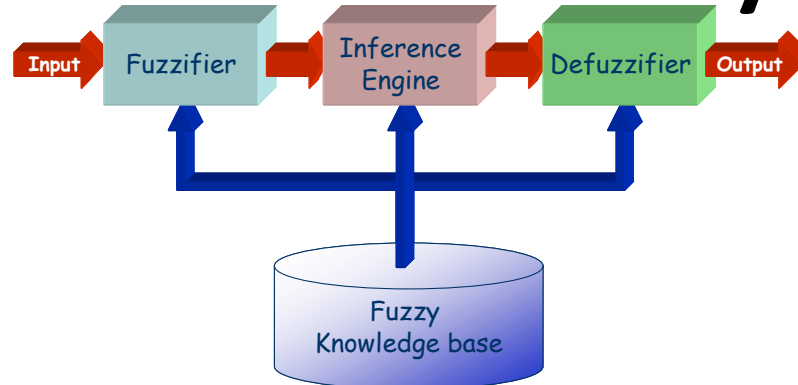
Because of its multi-disciplinary nature, the fuzzy inference system is known by a number of names, such as
fuzzy-rule-based system,
fuzzy expert system,
fuzzy model,
fuzzy associative memory,
fuzzy logic controller,
and simply fuzzy system.

Fuzzy Inference Systems

The Architecture of
Fuzzy Inference Systems



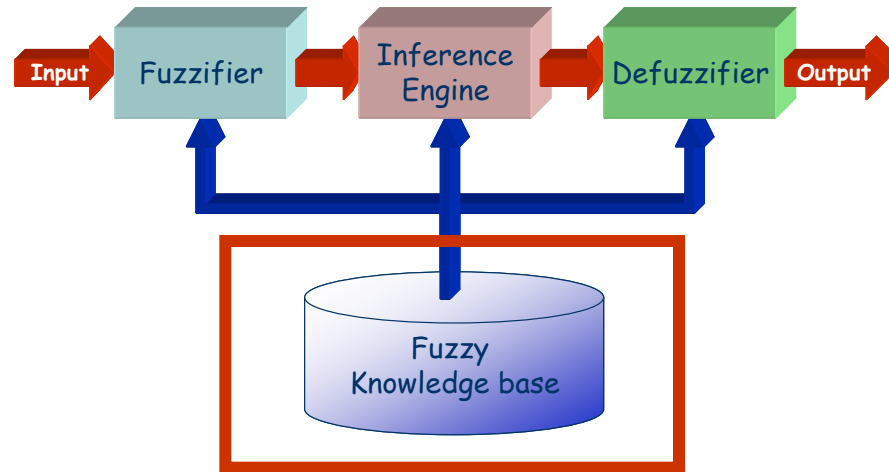
Fuzzy Inference Systems



The steps of *fuzzy reasoning* (inference operations upon fuzzy IF–THEN rules) performed by FISs are:

1. Compare the input variables with the membership functions on the antecedent part to obtain the membership values of each linguistic label. (this step is often called *fuzzification*.)
2. Combine (*usually multiplication or min*) the membership values on the premise part to get *firing strength* (*degree of fulfillment*) of each rule.
3. Generate the qualified consequents (either fuzzy or crisp) for each rule depending on the firing strength.
4. Aggregate the qualified consequents to produce a crisp output. (This step is called *defuzzification*.)

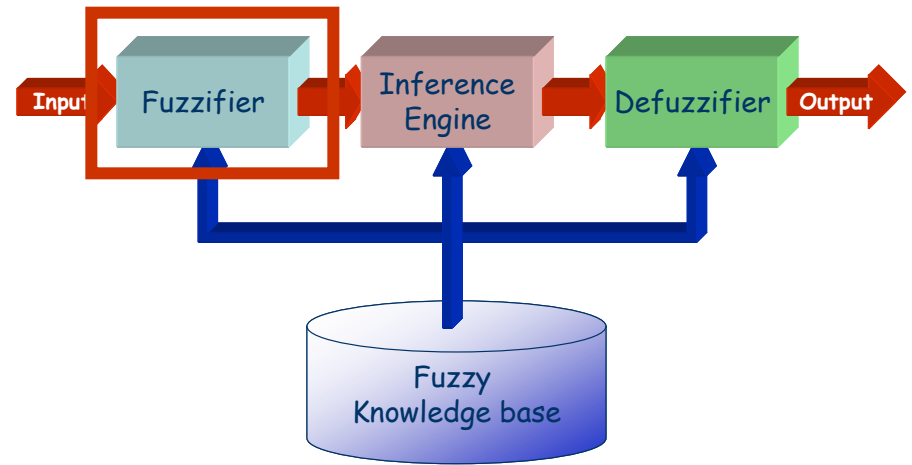
Fuzzy Knowledge Base



The **rule base** and the **database** are jointly referred to as the **knowledge base**.

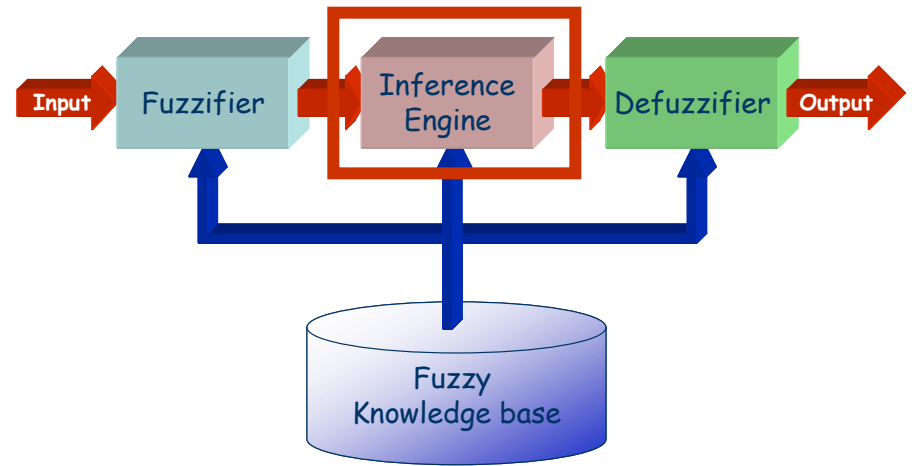
- a rule base containing a number of fuzzy IF–THEN rules;
- a database which defines the membership functions of the fuzzy sets used in the fuzzy rules

Fuzzifier

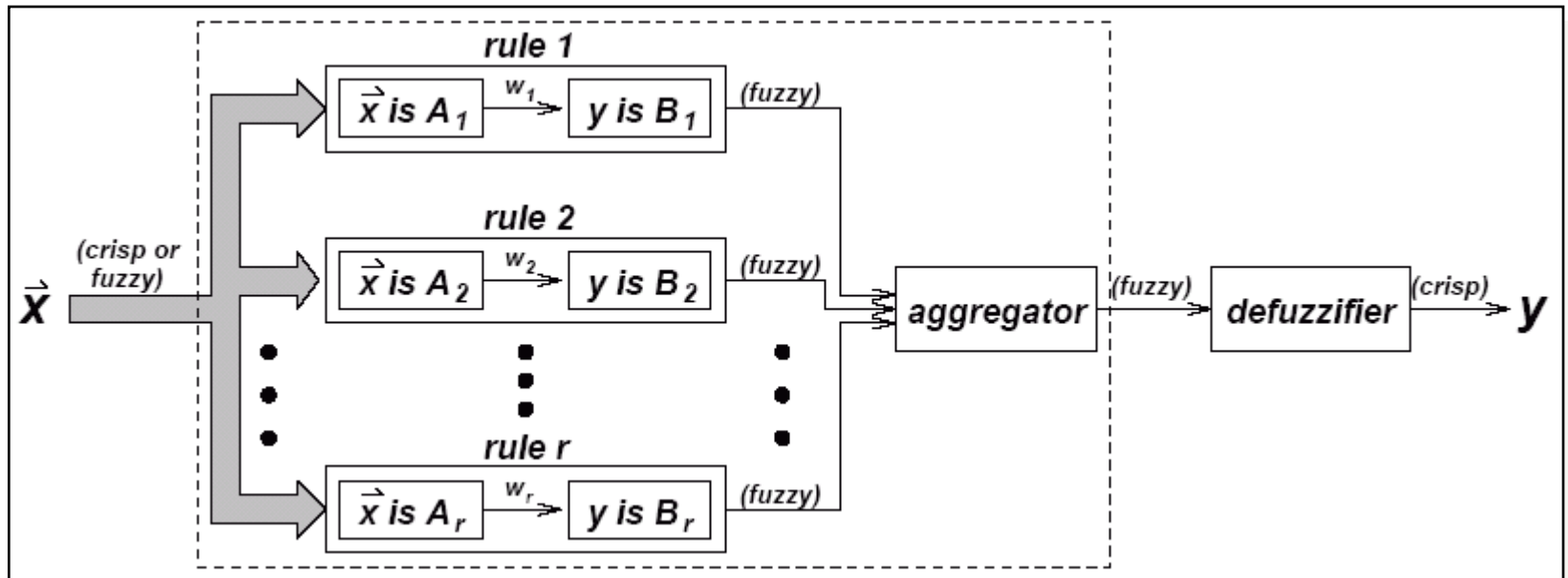


Converts the **crisp input** to a **linguistic variable** using the membership functions stored in the fuzzy knowledge base.

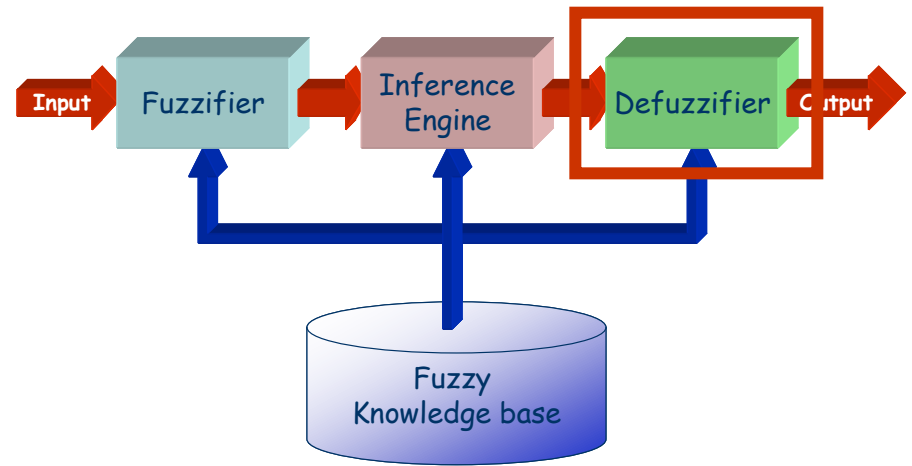
Inference Engine



Using **If-Then type fuzzy rules** converts the fuzzy input to the **fuzzy output**.

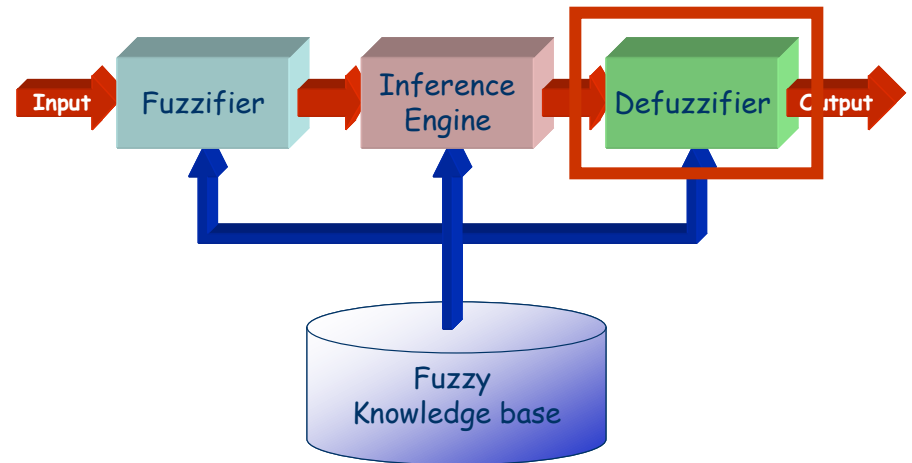


Defuzzifier



Converts the **fuzzy output** of the inference engine **to crisp** using membership functions analogous to the ones used by the fuzzifier.

Defuzzifier



- Converts the **fuzzy output** of the inference engine **to crisp** using membership functions analogous to the ones used by the fuzzifier.
- Five commonly used defuzzifying methods:
 - Centroid of area (COA)
 - Bisector of area (BOA)
 - Mean of maximum (MOM)
 - Smallest of maximum (SOM)
 - Largest of maximum (LOM)

Fuzzy Inference Methods

The most important two types of fuzzy inference method are **Mamdani** and **Sugeno** fuzzy inference methods,

Mamdani fuzzy inference is the most commonly seen inference method. This method was introduced by Mamdani and Assilian (1975).

Another well-known inference method is the so-called **Sugeno or Takagi–Sugeno–Kang** method of fuzzy inference process. This method was introduced by Sugeno (1985). This method is also called as TS method.

The main difference between the two methods lies in the **consequent of fuzzy rules**.

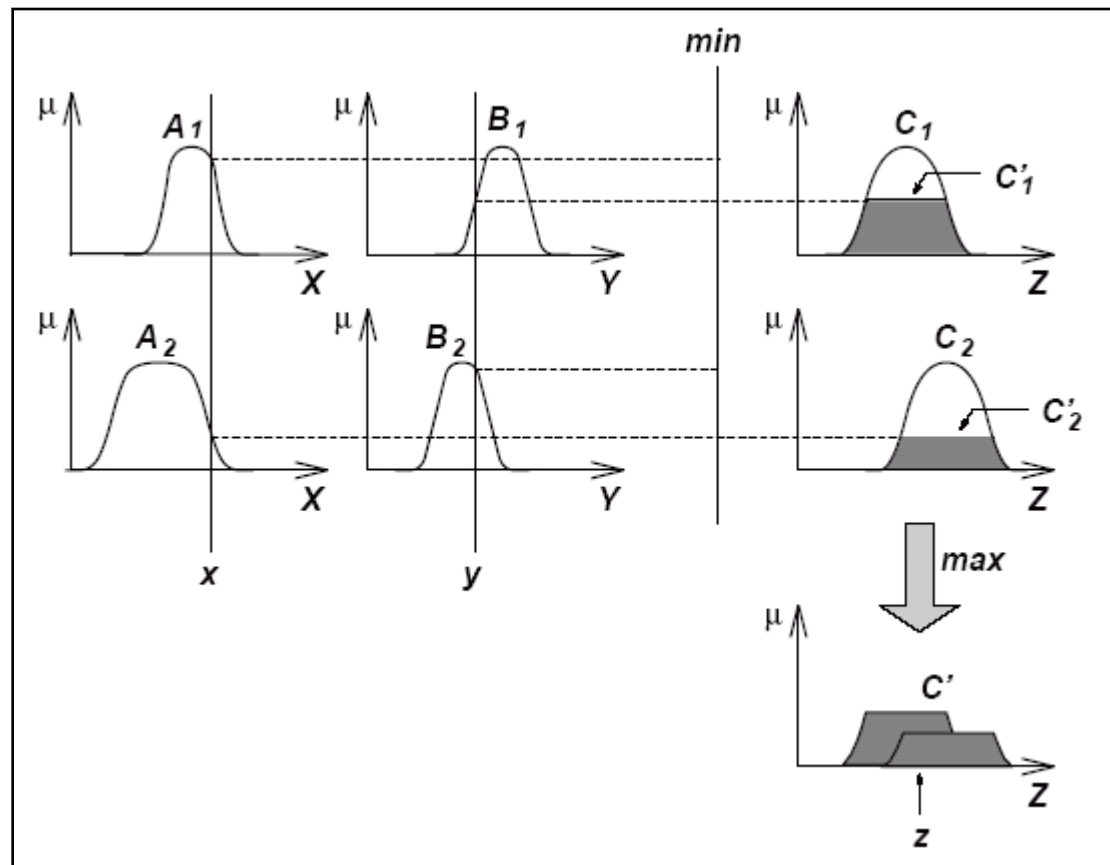
Mamdani Fuzzy models

To compute the output of this FIS given the inputs, six steps has to be followed

1. Determining **a set of fuzzy rules**
2. **Fuzzifying** the inputs using the input membership functions
3. **Combining the fuzzified inputs** according to the fuzzy rules to establish a **rule strength** (Fuzzy Operations)
4. Finding the **consequence of the rule** by combining the rule strength and the output membership function (implication)
5. **Combining the consequences** to get an output distribution (aggregation)
6. **Defuzzifying** the output distribution (this step is only if a crisp output (class) is needed).

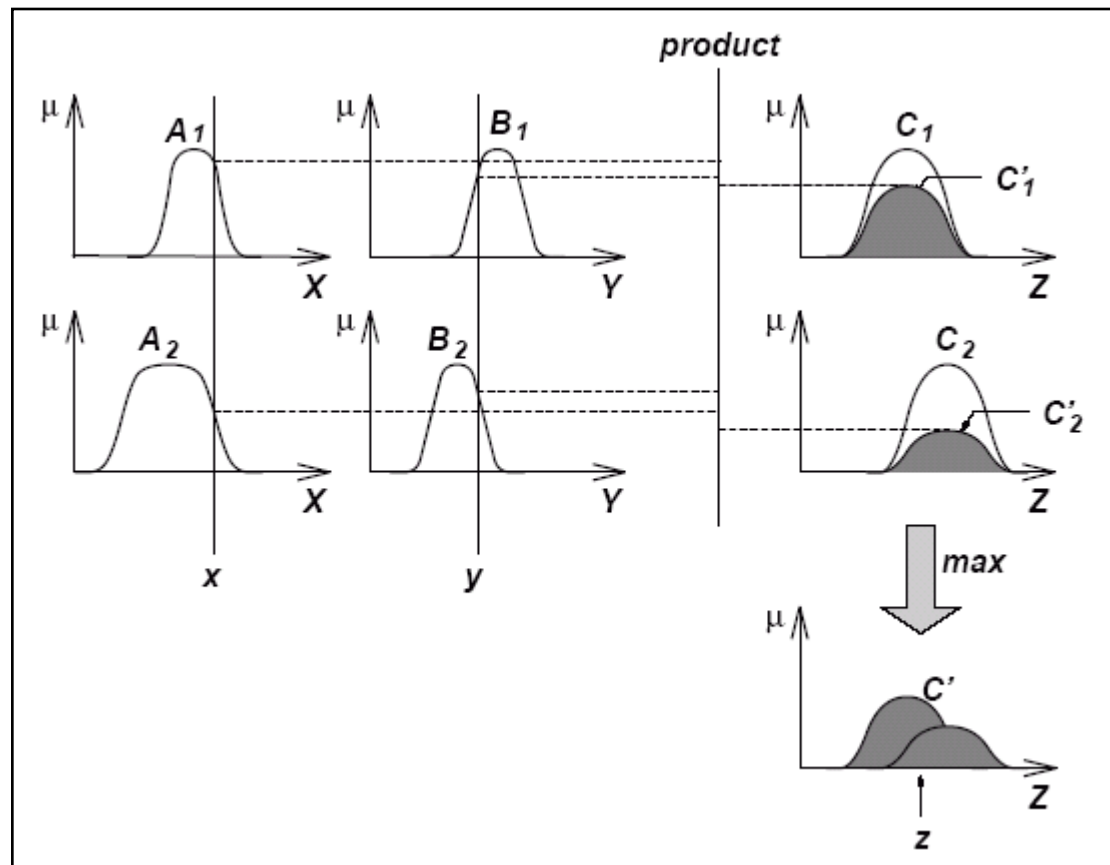
The Reasoning Scheme

Max-Min Composition is used.



The Reasoning Scheme

Max-Product Composition is used.



Sugeno Fuzzy Models

- Also known as **TSK fuzzy model**
 - Takagi, Sugeno & Kang
- Goal: **Generation of fuzzy rules** from a given input-output data set.

Fuzzy Rules of TSK Model

If x is A and y is B then $z = f(x, y)$

Fuzzy Sets

Crisp Function

$f(x, y)$ is very often a polynomial
function

Examples

R1: if X is small and Y is small then $z = -x + y + 1$

R2: if X is small and Y is large then $z = -y + 3$

R3: if X is large and Y is small then $z = -x + 3$

R4: if X is large and Y is large then $z = x + y + 2$

The Reasoning Scheme

